

Are Kalshi Prices Accurate Probabilities?

Oct 9, 2026 · @agasthya

Across 1,066 settled Kalshi markets, prices 24 hours before close were very well calibrated (calibration error 0.002), showed no favorite-longshot bias, and grew more accurate as markets neared close. Simple strategies betting against perceived mispricing lost money out of sample, mostly because of the bid-ask spread.

Question and data

Kalshi is a CFTC-regulated US exchange where people trade yes/no contracts on real events. A YES contract pays \$1 if the event happens and \$0 if not, so its price can be read as the market's probability. This study asks whether those probabilities are accurate, and whether any inaccuracy could be traded profitably.

Data. Settled markets were downloaded from Kalshi's public API ([historical data docs](#)). Each market's hourly bid and ask were recorded 1 hour, 24 hours and 7 days before close, together with its final YES/NO result.

Item	Value
Markets collected	3,000, from 437 randomly ordered series
Markets in the 24-hour analysis	1,066
Events those markets belong to	376
Overall YES rate vs average price	38.6% vs 37.1%
Categories with 50+ markets	Sports 551, Mentions 166, Entertainment 130, Financials 50
Excluded	Crypto markets; markets open under 30 hours; multi-leg combo markets

Method

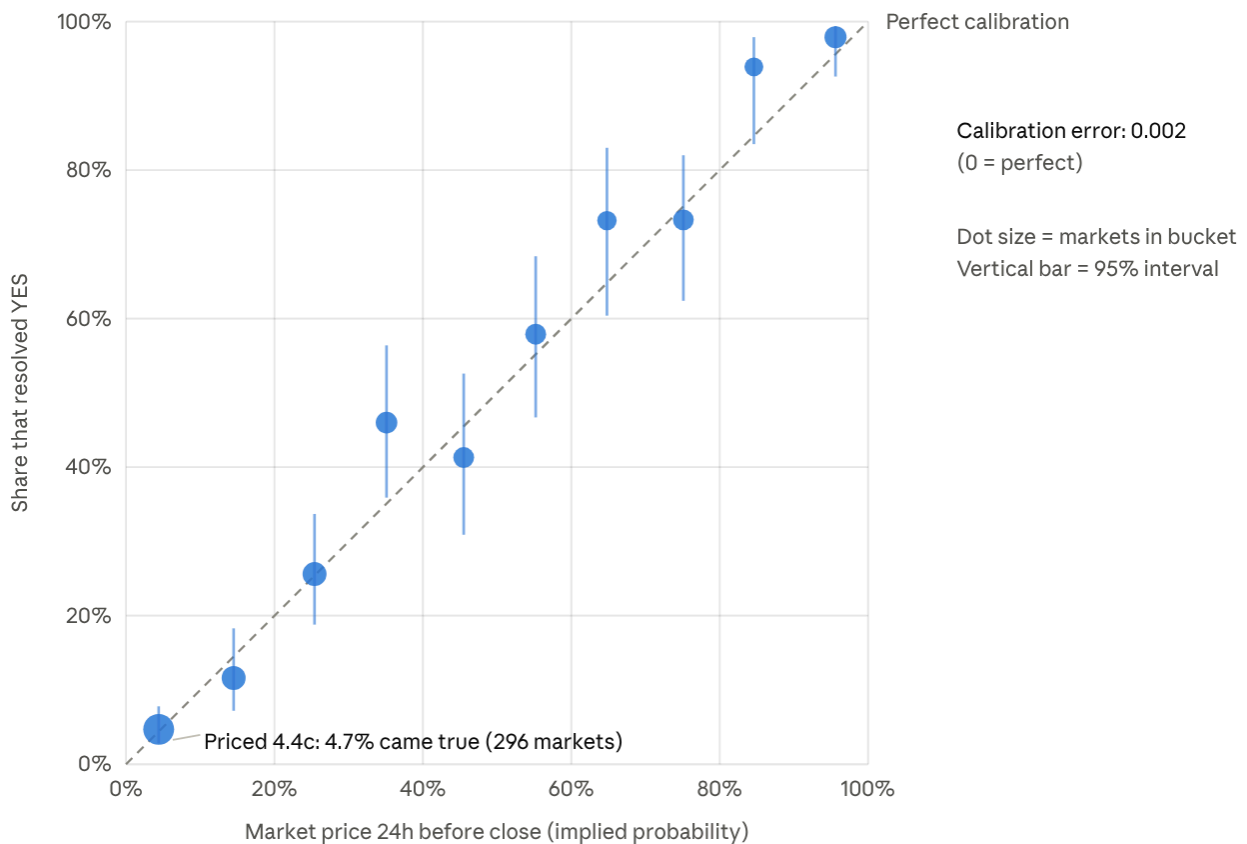
- Price.** The implied probability is the mid price, $(\text{bid} + \text{ask}) / 2$, at each snapshot time. The last trade was not used because it can be hours stale.
- Filters.** Snapshots with a missing bid or ask, a spread above 10¢, or market volume under 10 contracts were dropped.

3. **Calibration.** Markets were grouped into ten 10¢ price buckets. Each bucket's YES frequency was compared with its average price, with 95% Wilson confidence intervals.
4. **Scores.** Brier score, its Murphy decomposition (reliability, resolution, uncertainty) and log loss.
5. **Bias test.** A logistic recalibration, $P(\text{YES}) = \text{sigmoid}(a + b \times \text{logit}(\text{price}))$. A slope b above 1 indicates a favorite-longshot bias.
6. **Dependence.** Markets in the same event are linked (one CPI threshold resolving YES implies others do). All confidence intervals use a cluster bootstrap that resamples whole events, 1,000 draws.
7. **Tradability.** Two rules were backtested: buy NO when YES is cheap (fade longshots), and buy YES when YES is expensive (back favorites). Trades pay the ask and Kalshi's taker fee, $\text{round_up}(0.07 \times \text{contracts} \times P \times (1 - P))$. The threshold was chosen on the oldest 60% of markets and judged on the newest 40%.

Finding 1: prices are very well calibrated

Kalshi prices track real outcomes closely

1,066 markets, price 24 hours before close, grouped into 10-cent buckets



step3_calibration.py output · 1,066 settled Kalshi markets, 24h before close

Nine of ten buckets' 95% intervals cross the diagonal. The 30–40¢ bucket misses it narrowly (46% came true at a 35% price, 87 markets); with ten buckets, one such miss is roughly what chance alone produces. The Brier score was 0.1328 (95% CI 0.1185 to 0.1489) against 0.25 for always guessing 50%, and log loss 0.412 against 0.693. In the Murphy decomposition, calibration error (reliability) was 0.0021 while resolution was 0.1057: prices carry a lot of real information and are only very slightly off.

Finding 2: no favorite–longshot bias

Cheap contracts were not overpriced. In horse racing and sports betting, longshots famously win less often than their odds imply; on Kalshi the longshot gap was exactly zero.

Group (price 24h before close)	Markets	Actual YES rate – price	95% CI (event bootstrap)
Longshots (under 15¢)	366	+0.000	–0.022 to +0.026
Middle (15¢–85¢)	583	+0.020	–0.014 to +0.053
Favorites (over 85¢)	117	+0.035	–0.010 to +0.064

The recalibration slope was $b = 1.085$ (95% CI 0.967 to 1.247). The interval includes 1.0, so perfect calibration cannot be ruled out. The intercept $a = +0.145$ hints that YES was slightly underpriced overall, consistent with the 38.6% YES rate against a 37.1% average price, but no group gap is significant.

Finding 3: accuracy improves as markets approach close

On the same 176 markets priced at all three snapshot times, the Brier score fell from 0.1015 a week out to 0.0897 an hour before close.

Time before close	Brier score (lower = better)	Calibration error
7 days	0.1015	0.0062
24 hours	0.0963	0.0033
1 hour	0.0897	0.0059

Calibration error stays tiny at every horizon, so the gain comes from new information arriving, not from a bias being corrected: a week out the market is just as honest, it simply knows less.

A first version compared all markets available at each horizon (733, 1,066 and 571) and showed the 7-day prices as more accurate than the 24-hour ones. That was an artifact of comparing different sets of questions; restricting to the same markets removed it.

Finding 4: no tradable edge after costs

Both rules lost money on the 426 newest markets they had never seen. Training covered 640 markets closing up to 3 September 2026.

Rule (threshold chosen in training)	Test trades	Test events	Win rate	Profit (100 contracts per trade)	Return on capital	t-stat
Fade longshots, YES under 3¢	28	15	89.3%	-\$268.29	-9.70%	-1.67
Back favorites, YES over 80¢	57	32	93.0%	-\$86.83	-1.62%	-0.37

The bid-ask spread is the main cost, not the fee. Fading every longshot under 10¢ across all 1,066 markets:

Step	Amount
Profit if trades filled at the mid price	-\$118.25
Half-spread paid	-\$373.45
Kalshi fees	-\$61.68
Actual profit	-\$553.38

The spread cost about six times the fees. The strategy also lost before any cost, which matches Finding 2: there was no longshot overpricing to bet against.

Finding 5: the in-sample winner failed out of sample

"Back favorites" looked like a strong strategy on the training data and lost money on new data.

Back favorites, threshold 0.20	Training (older 640 markets)	Test (newer 426 markets)
Trades	86	57
Win rate	98.8%	93.0%

Back favorites, threshold 0.20	Training (older 640 markets)	Test (newer 426 markets)
Return on capital	+6.09%	-1.62%
t-stat (per event)	+4.10	-0.37

Training runs at several thresholds showed 100% win rates and t-statistics of 6 to 9. Those numbers come from picking the best of seven thresholds on a small sample with few losses, not from a real edge. Choosing the rule on old data and judging it only on newer data exposed this. Without the time split the study would have reported a profitable strategy.

Limitations

- **Sports-heavy sample.** Sports and Mentions make up 67% of the 24-hour sample; economics, politics and weather barely appear. The results describe those categories, not all of Kalshi.
- **Modest sample size.** 1,066 markets in 376 events; the time comparison uses 176 markets and the out-of-sample tests 28 and 57 trades. Several intervals are wide.
- **Brier scores are not comparable across groups.** They depend on how hard each group's questions are. Calibration error is the fair cross-group measure.
- **Hourly prices.** The snapshot is the last hourly candle before each target time, up to 6 hours stale.
- **Fills assumed.** The backtest assumes 100 contracts fill at the best ask; thin books would make real costs higher.
- **One fee schedule.** The general 0.07 taker formula was used; some products have their own fees and resting orders can be cheaper.

Next steps

- ☐ Rerun collection excluding Sports to get a fair sample of economics, weather and politics, then compare calibration across categories
- ☐ Raise the sample to 8,000+ markets to tighten the intervals
- ☐ Add a paired bootstrap for the Brier difference between horizons, to test whether the improvement is significant
- ☐ Test a maker strategy (resting limit orders instead of crossing the spread), since the spread was the dominant cost
- ☐ Compare against Polymarket on matched events to test cross-venue calibration

Code and how to reproduce

Four Python scripts, run in order. Settings sit at the top of each file.

Code: GitHub link to be added.

Script	What it does	Output
step1_collect.py	Downloads series, settled markets (series by series, up to 25 each) and hourly candlesticks	data/markets_raw.csv, data/snapshots_raw.csv
step2_clean.py	Joins, filters and computes mid prices	data/analysis.csv
step3_calibration.py	Calibration table, scores, bias test, breakdowns, charts	results/summary.txt, figures/
step4_backtest.py	Fee- and spread-aware backtest with a time split	results/backtest.txt

Dependencies: requests, pandas, numpy, matplotlib. No account or API key is needed; only public market data is read.